DO NOW — RECALL

Processors, scheduling and LMC practice

Name: _____ Class: _____ Date: _____

1. What does LMC STA do?

2. What is the role of MAR?

3. What is a time quantum?

4. Full adder 1+1+0: sum and carry bits?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Copies ACC contents to the specified mailbox.

2. Model answer

Address of the memory location being accessed.

3. Model answer

Maximum CPU slice before RR rescheduling.

4. Model answer

10_2: sum0, carry1.

By the end, you can…

• Trace the LMC accumulator and mailbox separately: INP overwrites ACC, while STA copies to storage.

• Schedule independent CPU jobs with an explicit arrival/time rule and label every Gantt boundary.

• Distinguish processor speed, memory cache and interrupt/OS services when explaining performance.

Vocabulary — know these terms

ACC — Accumulator register.

FCFS — First Come First Served.

SJF — Shortest Job First.

pipeline — Overlap instruction stages.

NEW LEARNING

Read, annotate and explain

Understand

Trace the LMC accumulator and mailbox separately: INP overwrites ACC, while STA copies to storage.

Apply

Schedule independent CPU jobs with an explicit arrival/time rule and label every Gantt boundary.

Evaluate

Distinguish processor speed, memory cache and interrupt/OS services when explaining performance.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace sequential instruction fetch when PC=90.

Predict first

PC holds address of next instruction, not the instruction contents.

Explain and check

Explain the reason for the result.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Using LMC code above, trace ACC after each input, LDA, SUB and OUT. [6]

Question 2

Processes A (burst 4), B (burst 2) and C (burst 1) all arrive at t=0. For FCFS, use the stated arrival-list order A, B, C to break ties; for non-preemptive SJF choose the shortest available job. Compare execution orders and mean waiting times. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate clock speed, cache and pipelining for a task with heavy memory latency and branches. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

LMC STA X loads ACC from mailbox X.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.